Practical 11(Euler’s Method)

Question 1

```
EulerMethod[a0_, b0_, n0_, f_, alpha_] :=
  Module[{a = a0, b = b0, n = n0, h, ti},
   h = (b - a) / n;
   ti = Table[a + (j - 1) h, {j, 1, n + 1}];
   ui = Table[0, {n + 1}];
   ui[[1]] = alpha;
   OutputDetails = {{0, ti[[1]], alpha}};
   For[i = 1, i ≤ n, i++,
    ui[[i + 1]] = ui[[i]] + h * f[ti[[i]], ui[[i]]];
    OutputDetails = Append[OutputDetails,
      {i, N[ti[[i + 1]]], N[ui[[i + 1]]]}];
   ];
   Print[NumberForm[
     TableForm[OutputDetails, TableHeadings → {None, {"i", "ti", "ui"}}], 6]];
   Print["Subinterval size h used =", h];
  ];
f[t_, w_] := 1 + w / t;
a = 1; b = 6;
n = 10;
alpha = 1;
EulerMethod[a, b, 10, f, alpha];
```

i	ti	ui
0	1	1
1	1.5	2.
2	2.	3.16667
3	2.5	4.45833
4	3.	5.85
5	3.5	7.325
6	4.	8.87143
7	4.5	10.4804
8	5.	12.1448
9	5.5	13.8593
10	6.	15.6193

Subinterval size h used $=\frac{1}{2}$

Question 2

```
EulerMethodwithH[a0_, b0_, h0_, g_, alpha_] :=
  Module[{a = a0, b = b0, h = h0, n, ti},
   n = (b - a) / h;
   ti = Table[a + (j - 1) h, {j, 1, n + 1}];
   ui = Table[0, {n + 1}];
   ui[[1]] = alpha;
   OutputDetails = {{0, ti[[1]], alpha}};
   For[i = 1, i ≤ n, i++,
    ui[[i + 1]] = ui[[i]] + h * f[ti[[i]], ui[[i]]];
    OutputDetails = Append[OutputDetails,
      {i, N[ti[[i + 1]]], N[ui[[i + 1]]]}];
   ];
   Print[NumberForm[
     TableForm[OutputDetails, TableHeadings → {None, {"i", "ti", "ui"}}], 6]];
   Print["Subinterval size h used =", h];
  ];
g[t_, w_] := 1 + w / t;
a = 1; b = 6;
h = .2;
alpha = 1;
EulerMethodwithH[a, b, h, g, alpha];
```

i	ti	ui
0	1.	1
1	1.2	1.2
2	1.4	1.44
3	1.6	1.728
4	1.8	2.0736
5	2.	2.48832
6	2.2	2.98598
7	2.4	3.58318
8	2.6	4.29982
9	2.8	5.15978
10	3.	6.19174
11	3.2	7.43008
12	3.4	8.9161
13	3.6	10.6993
14	3.8	12.8392
15	4.	15.407
16	4.2	18.4884
17	4.4	22.1861
18	4.6	26.6233
19	4.8	31.948
20	5.	38.3376
21	5.2	46.0051
22	5.4	55.2061
23	5.6	66.2474
24	5.8	79.4968
25	6.	95.3962

Subinterval size h used =0.2

Question 3

```
f[t_, x_] := x;
h = 0.2;
EulerMethodwithH[0, 0.4, h, f, 1];
```

i	ti	ui
0	0.	1
1	0.2	1.2
2	0.4	1.44

Subinterval size h used =0.2

Question 4

```
f[t_, x_] := x;
n = 2;
EulerMethod[0, 0.4, n, f, 1];
```

i	ti	ui
0	0	1

Subinterval size h used =2

Question5

```
DSolve[{x'[t] == x[t], x[0] == 1}, x[t], t]
```

$\{\{x[t] \to e^{t}\}\}$

```
Abs[1.2 - N[Exp[0.2]]]
```

0.0214028

```
Abs[1.44 - N[Exp[0.4]]]
```

0.0518247